LYRA-CPP — THETIS DIRECT-PORT PLAN (TX)

Authoritative doc for the current TX work direction.

> *[!] The older `EXECUTION_PLAN.md` (Step 14 wire-layer rebuild, Phase 3 etc.) describes a DIFFERENT approach that is now superseded for the TX work. Do NOT mix the two plans — that doc tracks an earlier wire-rebuild pass which was partial and Lyra-native-leaning. THIS doc is the active plan going forward. Old doc remains in tree for historical reference only (some shipped pieces from it are still in production — noted per-component in the registry below).*

Branch: `tx-rebuild` (the direct-port work lives here; `origin/main` carries the earlier Step 14 / TX-1 line and is deliberately untouched by the direct-port arc).

Reference tree (the ONLY trusted TX reference):

- `D:\sdrprojects\OpenHPSDR-Thetis-2.10.3.13\Project Files\Source\`
    - `wdsp\*.c` — WDSP DSP engine (GPL v3+, port-with-attribution OK)
    - `ChannelMaster\*.c` — wire layer (GPL v3+, port-with-attribution OK)
    - `Console\*.cs` — C# UI/control glue (study ONLY; write Lyra-native equivalents)
- `Y:\Claude local\_hl2src\` — HL2+ ak4951v4 gateware RTL

(`control.v` / `radio.v` / `usopenhpsdr1.v` etc. — the ground truth when host-side reference and gateware disagree)

Provenance & attribution rule (operator-locked 2026-06-09, supersedes the 2026-05-31 "no-attribution-in-commits" tightening):

Lyra-cpp is a derivative work of openHPSDR Thetis (ChannelMaster + WDSP modules), GPL v3+. The GitHub project description openly states this: *"TX baseline ported from openHPSDR Thetis (ChannelMaster) with Lyra-native DSP additions."* Open attribution is both legally required (GPL) and operator-intended (public project posture).

1. **License attribution at file head — REQUIRED.** Any file ported from WDSP or ChannelMaster carries a header block crediting upstream (NR0V for WDSP modules; openHPSDR Thetis for ChannelMaster modules) + the GPL v3+ notice. Existing `AAMix.{h,cpp}`, `CMaster.cpp`, `RadioNet.cpp` etc. already follow this; keep it.
2. **Code comments citing source — ENCOURAGED.** Citations may name the reference openly (`Reference: Thetis ChannelMaster cmaster.c:297-313`, `// Per openHPSDR networkproto1.c:1078`, `// HL2+ ak4951v4 gateware control.v:209-220`). The file:line is the primary technical content; the reference-app name is context that helps future readers find the source tree.
3. **Commit messages — REFERENCE-APP NAMES PERMITTED.** A commit like *"port aamix.c from Thetis ChannelMaster with Lyra-native idiom translations"* is fine — it matches the public GitHub posture and makes the history grep-able. The earlier `4cef88d` tightening (2026-05-31) that put commit messages on the no-attribution list is **rescinded**.
4. **Public docs (README, `docs/*.md` design docs, memory files, this plan) — OPEN ATTRIBUTION.** Same posture as the GitHub description.
5. **User-facing UI strings — KEEP OPERATOR-FOCUSED.** UI labels operators see in normal operation (panel titles, button text, error toasts) describe Lyra behavior, not provenance. *"AAMix output (HL2 jack)"* not *"Thetis-ported AAMix output"*. About dialog / credits screen / install docs MAY (and probably should) name openHPSDR Thetis + WDSP/NR0V openly — credit where due. This is the only category that stays "Lyra-native voice in operator-facing labels," because operators interact with Lyra, not its source provenance.
6. **The "Lyra-Native style governs surrounding architecture only" rule (Stage B-locked) STANDS unchanged.** Reference DSP API call patterns are byte-for-byte faithful; Qt/Vulkan/QML/process model is Lyra-native.
7. **Reference precedence STANDS unchanged.** Verify TX ONLY against Thetis 2.10.3.13 (`D:\sdrprojects\OpenHPSDR-Thetis-2.10.3.13\...`) + the HL2+ ak4951v4 gateware RTL (`Y:\Claude local\_hl2src\`). **Old**

Python Lyra is NOT a trusted TX reference (its TX never worked).

Consequence of the 2026-06-09 amendment: the three historical commits (`efdde02`, `604d87b`, `ed9f0cd`) that "predated the no-attribution rule" and contained "Thetis" in their messages are **no longer in violation** — they're now consistent with the rule. Task #73 ("Cleanup pre-existing 'Thetis' mentions in shipped code") is **OBSOLETED** by this amendment.

LOCKED METHODOLOGY (Stage B-validated, do NOT relax)

These rules were earned through the Stage B aamix.c port arc (2026-06-08) — see SESSION_LOG.md for the full trail.

1. **Smallest revertable empirical step -> operator HL2 bench ->

next step.** Each commit is one focused change with a falsifiable bench gate. Multi-step "design then implement" is for stages that genuinely need it (operator approves); default is iterate-and-bench.

2. **"Do as Thetis does, Lyra-Native style"** (locked

2026-05-30 after the `fexchange0` crash arc, re-validated tonight after the B.6.b shortcut attempt cost a revert cycle). Three sub-rules:

- Every WDSP API call matches the reference byte-for-byte

(parameter values, order, the SET of setters fired, the lifecycle stage at which they fire). If tempted to pick a value the reference doesn't use, STOP and find what the reference uses. "I have a reason to pick differently" means I don't. Reference has been right every time.

- "Lyra-Native style" governs surrounding architecture ONLY

(Qt + Vulkan + QML + process/thread model + facade APIs + IPC + UI shell). The WDSP DSP-engine call surface is reference-bound.

- Provenance only in code COMMENTS + design docs + memory.

Never commit messages, never UI strings.

3. **Counters first when the symptom is "nothing happens".**

The Stage B 2-stage instrumentation (chain counters in xMixAudio/mix_main + dispatch counters in dispatchAudioFrame) localized the silent-audio defect to a single line of code on the 2nd bench. Pure speculation would have cost a multi-round revert/speculate cycle. Atomic counter + rate-limited qInfo = trivial overhead, very high diagnostic ROI when "I press play, hear nothing, log looks normal."

4. **Operator-empirical rule overrides agent inference.**

When operator hardware data contradicts what an audit/red-team/Plan-agent concluded, the data wins. Do not defend the audit — re-open, bisect, fix. This rule has fired repeatedly through the broader project arc; the Stage B B.6.b-fix1 fix was an instance (chain counters healthy suggested AAMix was correct; the temptation was to revert the whole port; the empirical pattern instead pointed at the wire-up surface where the bug actually was).

5. **Read the reference FIRST, in shipped-code form, before

designing.** Every time we designed first and read second, we shipped a regression. Every time we read first, we got the right answer in one pass. The no-attribution rule applies to shipped code; it does NOT mean "don't read the reference." Read it, dossier the file:line in the plan doc, then write Lyra-cpp.

6. **Session-start discipline:** `git fetch origin` is the

FIRST action every session, before reading any local state. If divergence exists, surface it to the operator BEFORE writing code. No exceptions.

7. **Push after every component ships, not at EOD.** Each

clean operator HL2 bench -> backup bundle + push immediately. Smaller in-flight delta = smaller surprise blast radius.

8. **Backup bundle naming:** `_backups/lyra-cpp-YYYY-MM-DD-.bundle`.

`git bundle create … --all` (covers all refs).

STATUS REGISTRY — what's done, in-progress, pending

Format: per-file or per-component row. Status tags (ASCII so they render identically in the PDF/DOCX siblings):

• [DONE] DONE + operator-bench-validated

• [WIP] IN-PROGRESS (committed but not yet bench-validated, or partially shipped)

• [PEND] PENDING (planned, not started)

• [N/A] N/A (Lyra-native equivalent already shipped; no direct port planned — see notes column)

Wire layer (ChannelMaster — port directly w/ attribution)

Source	Target (lyra-cpp)	Status	Notes
WDSP exports linkage (`OpenChannel`/`fexchange0`/analyzers/PS family)	`src/wire/wdspcalls.{h,cpp}` (P0.a)	[DONE]	**P0.a** `e0c0f4a` **2026-06-09.** The single operator-approved linkage seam: fn-ptr table carrying the WDSP exports' exact names, resolved once from the loaded wdsp.dll. PureSignal entry family complete (verified vs dumpbin). RESAMPLE-typed + `flush_resample` added at P0.c.
`ChannelMaster\cmcomm.h` (family common header)	`src/wire/cmcomm.{h,cpp}` (P0.b -> P0.d)	[DONE]	`complex` typedef + PORT + verbatim malloc0 (wdsp/utilities.c:37-43) + PI + the reference include surface. P0.d added the opaque verbatim twin typedefs for deferred-subsystem types (ANB/NOB/VOX/TXGAIN/ANALYZERS + DELAY at P2.a); **EER completed at P2.a 2026-06-12** — full verbatim struct (wdsp/eer.h:30-49) + the umbrella-mapping note (.cpp files include explicit family headers — an include-list umbrella would be circular under `#pragma once`).
`ChannelMaster\ilv.c` (TX I/Q interleaver)	`src/wire/ILV.{h,cpp}` (P0.b)	[DONE]	**P0.b** `1f49d98` **2026-06-11.** VERBATIM direct port (`ilv`, `*ILV` twin typedef, raw Outbound fn ptr, `pcm->xmtr[].pilv` setters, malloc0/_aligned_free). Mechanical diff vs ilv.{h,c} = whitespace-only. The earlier `src/wdsp/ILV` retrofit + its pilv[] bank are retired. scratch/test_ilv ALL PASS.
`ChannelMaster\cmbuffs.{h,c}` (CMB ring + cm_main pump)	`src/wire/CmBuffs.{h,cpp}` (P0.d)	[DONE]	**P0.d** `afc7950` **2026-06-12.** VERBATIM rewrite (`cmb`, `*CMB`, `#define CMB_MULT (3)`, malloc0/_aligned_free; the Phase-C retrofit's calloc/intptr_t/guard deviations removed; `pcm->in[]` alloc moved back to create_cmaster per reference). Mechanical diff = whitespace-only. [DONE] **P0.d operator HL2 RX-regression bench PASSED 2026-06-12.**
`ChannelMaster\cmsetup.{h,c}` (radio structure + id helpers)	`src/wire/cmsetup.{h,cpp}` (P0.d, NEW)	[DONE]	**P0.d** `afc7950`**.** VERBATIM: cmMAX* sizing macros (16/4/4/2/32), rxid/txid/sp0id/stype/chid/inid/mixinid/getbuffsize, SetRadioStructure + set_cmdefault_rates. main.cpp config: SetRadioStructure(2,1,1,1,0,…) -> chid(0,0)=0 RX1 / chid(1,0)=1 TX matches the live channel layout. [DONE] Bench PASSED 2026-06-12.

Source	Target (lyra-cpp)	Status	Notes
`ChannelMaster\cmaster.{h,c}` (FULL: struct + create/destroy + xcmaster + setters)	`src/wire/CMaster.{h,cpp}` (P0.d)	[DONE]	**P0.d** `afc7950`**.** VERBATIM rewrite: full `_cmaster` struct (cmaster.h:39-99 incl. the PS surface — out[3]/panalalloc/pgain/peer), `cmaster`, `*CMASTER`, raw TCI callback fn ptrs, `enum AudioCODEC`, `cm = {0}`. Verbatim no-arg create_xmtr/destroy_xmtr (out[0..2] malloc0 + OpenChannel(ch 1) + XCreateAnalyzer(TX disp 1) + create_ilv through the wdspcalls seam — **the TxChannel RAII carve-out is DELETED**, `src/wdsp/TxChannel.{h,cpp}` retired). create_cmaster/destroy_cmaster verbatim per-stream loops (update[] CS + cmbuffs + in[] for all cmSTREAM streams). xcmaster verbatim (update[] CS restored, real stype/txid/chid, TCI memset restored; fexchange0 + monitor xMixAudio + xilv live). All Sendp*/Set* setters ported. Deferred subsystem lines (dexp/anti-vox/txgain/sidetone/pipe/cmasio/analyzers.c/create_rcvr-body) carried IN PLACE as reference text with DEFERRED tags; **eer RESTORED VERBATIM at P2.a 2026-06-12** (create_eer run=0 / destroy_eer / xeer / pSetEER* via the wdspcalls seam — the object exists so the sendOutbound/sendProtocol1Samples `peer->run` derefs are valid, per cmaster.c:212-224). Sole code accommodation: `(char*)""` at XCreateAnalyzer. [DONE] **Operator HL2 RX-regression bench PASSED 2026-06-12** (RX working on the new per-stream pump/ring layout).
`ChannelMaster\obbuffs.c` (TX-out ring/seam)	`src/wire/ObBuffs.{h,cpp}` (P1, NEW)	[WIP]	**P1 SHIPPED 2026-06-12** — obbuffs.{h,c} verbatim (obb/*OBB twin typedef, file-scope obp four-alias bank, the reference's own calloc/free 2014-era idioms kept). Mechanical diff: .cpp IDENTICAL; .h sole delta = the include-guard define replaced by #pragma once (documented packaging). Sole deferred line: `sendOutbound(id, a->out)` in ob_main (network.c:1237 = P2 scope), carried as reference text. TU is DORMANT — no Lyra caller of create_obbuffs until P3/P4; wire bytes unchanged.
`ChannelMaster\networkproto1.c` (HL2 wire writer)	`src/hl2_stream.cpp` (LIVE) + `src/wire/*` (Step 14 pieces)	[WIP]	This is the file the Step 14 plan was attacking. Currently a MIX of Step 14 Lyra-native rewrites + the original `hl2_stream.cpp` body. P2 = sendOutbound/sendProtocol1Samples fidelity audit of the dormant wire layer; P4 = Wire-LIVE switchover (one commit, full HL2 bench gate).
`ChannelMaster\network.c` (UDP + keepalive)	`src/wire/Ep2SendThread.{h,cpp}` + `src/wire/Ep6RecvThread.{h,cpp}`	[WIP]	Step 14 Lyra-native rewrites shipped (§5/§6); could be re-ported direct from reference, OR left as-is if reference-parity audit passes. Operator call (folds into P2/P4).

Source	Target (lyra-cpp)	Status	Notes
`ChannelMaster\netInterface.c::create_rnet`	`src/wire/RadioNet.cpp::create_rnet` (LIVE)	[DONE]	Direct port shipped earlier. The Stage-A SendpOutboundRx no-op stub at line 272 was DELETED 2026-06-08 (B.6.b-fix1, commit `8b8e0da`) because it clobbered the AAMix wire-up; future codec-id switching must NOT re-introduce a default registration here. P3 = the outbound registrations (SendpOutboundTx(OutBound) etc.) — MUST register AFTER create_xmtr (the verbatim setters have NO null guards, reference ordering).

WDSP RX chain (port directly w/ attribution)

Source	Target (lyra-cpp)	Status	Notes
`ChannelMaster\aamix.{c,h}`	`src/wire/AAMix.{h,cpp}` (P0.c)	[DONE]	**P0.c** `e0f2584` **2026-06-11 — VERBATIM re-port** (supersedes the 2026-06-08 Stage-B C++23-idiom translation, which the operator rejected as a deviation class). `aamix, *AAMIX` twin typedef + anonymous slew struct, raw Outbound fn ptr, paamix[] bank private to the .cpp, Win32 primitives direct. Mechanical diff vs aamix.{h,c} = whitespace-only. `src/wire/resample.h` = the public RESAMPLE ABI verbatim. WdspEngine RX wire-up: capturing lambda -> static member fn + TU-scope context ptr (the reference free-fn+global shape). [DONE] **Operator HL2 RX bench PASSED 2026-06-11.**
`wdsp\firmin.c` (filter cores)	(consumed via WDSP cffi)	[N/A]	Used internally by WDSP RXA chain via cdef + dlsym. No source-level port needed (the DLL ships with Lyra-cpp via `_native/` per the Python-project pattern; cdef + dlsym in `wdsp_native.cpp`). WISDOM cache fix shipped on Python side; lyra-cpp's RX chain inherits the same path.
`wdsp\wisdom.c`	(consumed via WDSP cffi)	[N/A]	Same as firmin.c.

WDSP TX chain (per CLAUDE.md §4.1 port table)

Source	Target (lyra-cpp)	Status	Notes
`wdsp\TXA.c` (channel scaffold)	(consumed via `wire/wdspcalls` seam — P0.a/P0.d)	[DONE]	**TxChannel class DELETED at P0.d** `afc7950` — the verbatim create_xmtr in `wire/CMaster.cpp` opens the WDSP TXA channel itself (OpenChannel/fexchange0/CloseChannel through the wdspcalls seam, cmaster.c:177-190 byte-exact), which removed the runtime-DLL RAII justification the class existed for. `wdsp_native.{h,cpp}` still carries the SetTXA*/GetTXAMeter cdefs for the operator-control surface.
`wdsp\bandpass.c` (bp0/bp1)	(consumed via WDSP cffi — SetTXABandpassFreqs)	[N/A]	The §15.23 trap (SetTXABandpassRun overrides bp1 to a stale state) is documented in CLAUDE.md + project_lyra_cpp_tx.md. cffi-only access is the correct posture.
`wdsp\compress.c` (TX compressor)	`src/wdsp/Compress.{h,cpp}` OR cffi binding	[PEND]	Operator choice: direct port (~150 LOC) for v0.2.1 EQ + dynamics ship, OR cffi-only binding if WDSP DLL exports are sufficient. Direct port recommended for parity with Lyra's existing UI control surface.
`wdsp\cfcomp.c` (5-band CFC)	`src/wdsp/CFComp.{h,cpp}` OR cffi binding	[PEND]	Same operator-choice. ~600 LOC; one of the heavier ports. Replaced in Python Lyra plan by a Lyra-native Combinator per §15.19; cpp may take the same path (custom Combinator instead of WDSP CFC) — operator design call.
`wdsp\osctrl.c` (CESSB)	cffi binding (SetTXAosctrlRun)	[PEND]	Likely cffi-only; ~200 LOC port not justified unless operator UI need surfaces.
`wdsp\wcpagc.c` mode 5 (leveler)	cffi binding (SetTXALeveler*)	[WIP]	cdef already present in `wdsp_native.cpp`. UI surface exists in TX-1 components on `main`. Confirm reference-parity at the call-site setter sequence.
`wdsp\wcpagc.c` mode 5 (ALC)	cffi binding (SetTXAALCMaxGain)	[WIP]	cdef present; SetTXAALCThresh does NOT exist (§15.23-class trap — only MaxGain governs ALC ceiling).
`wdsp\eqp.c` (parametric EQ)	`src/wdsp/EQ.{h,cpp}` OR cffi binding	[PEND]	Per CLAUDE.md §4.1 v0.2.1; ~300 LOC. Lyra-native parametric EQ likely (§15.19 lists 3-or-5-band parametric replacing the WDSP 10-band graphic).
`wdsp\gen.c` (gen0/gen1 generators)	cffi binding (SetTXAPreGen*/SetTXAPostGen*)	[WIP]	gen1 (postgen) used today for TUN tone. gen0 (pregen) cdefs deferred per §15.23 (bench-tooling nicety, not on the signal path).

TX PureSignal (per CLAUDE.md §4.1 v0.3)

Source	Target (lyra-cpp)	Status	Notes
`wdsp\iqc.{c,h}`	`src/wdsp/Iqc.{h,cpp}` (~315 LOC)	[PEND]	v0.3. cffi math + Lyra-cpp wrapper for the 5-state PS lifecycle (RUN, BEGIN, SWAP, END, DONE).
`wdsp\calcc.{c,h}`	`src/wdsp/Calcc.{h,cpp}` (~1164 LOC)	[PEND]	v0.3. cffi math + Lyra-cpp wrapper for the 8-state PS FSM + 3-state auto-attenuator FSM + PSDialog UI.
`wdsp\lmath.c::xbuilder`	`src/wdsp/PsXBuilder.{h,cpp}` (~200 LOC)	[PEND]	v0.3. Cubic-spline coefficient builder, used by Lyra-cpp-side calcc orchestration.
`wdsp\delay.c`	`src/wdsp/DelayLine.{h,cpp}` (~80 LOC)	[PEND]	v0.3. TX/feedback time-alignment.

Console (C# — study only, Lyra-native equivalents)

Source	Target (lyra-cpp)	Status	Notes
`Console\console.cs::chkMOX_CheckedChanged2`	`src/ptt.{h,cpp}` + `src/wdsp_engine.cpp` keydown/keyup	[WIP]	Lyra-native FSM shipped on `main`. Verified against the reference's keydown/keyup ordering invariants (RX-DSP stop on keydown -> wire MOX -> TX-DSP start; keyup TX-DSP off -> mox_delay -> clear MOX -> ptt_out_delay -> RX-DSP restart). No direct port — C# is study-only by license.
`Console\console.cs` (TR delays)	`src/tx/TrSequencing.{h,cpp}` (LIVE)	[DONE]	Lyra-native shipped with operator's Thetis-DB-verified defaults (mox=15/rf=50/space_mox=13/ptt_out=5/key_up=10 ms).
`Console\console.cs::m_bATTonTX`	`src/tx/AttOnTxPolicy.{h,cpp}` (LIVE)	[WIP]	Lyra-native shipped on `main`. Reference posture: keydown save per-band step-att + force-31 (when PS-A off OR CW); keyup restore. Implementation verified against `console.cs:30293-30327` + `:30391-30410`.
`Console\PSForm.cs`	`src/ui/PsDialog.{h,cpp}` (NEW, v0.3)	[PEND]	v0.3 PureSignal UI. C# study only; write Qt/QML native.
`Console\HPSDR\IoBoardHl2.cs`	`src/hl2_stream.cpp` (LIVE)	[WIP]	HL2 I/O quirks already implemented Lyra-native via the protocol-byte verification + gateware-RTL ground-truth pass (CLAUDE.md §15.26).

STAGE INDEX — ordered work plan

P0 verbatim-rewrite arc (2026-06-09 -> current; the ACTIVE plan)

After the operator rejected the 2026-06-09 "rule-#8" retrofit deviations ("I said PORT!"), every TX-ported file is being rewritten VERBATIM (byte-faithful; only documented packaging differences — `lyra::wire` namespace, PORT-expands-empty, the wdspcalls seam for the statically-linked WDSP exports). Each step ships with a mechanical diff vs the reference + an operator HL2 bench gate.

Step	Status	What	Commit / gate
P0.a	[DONE] DONE	`wire/wdspcalls.{h,cpp}` — the single approved WDSP linkage seam (exact export names, PS entry family complete)	`e0c0f4a` 2026-06-09

Step	Status	What	Commit / gate
P0.b	[DONE] DONE	ilv.{h,c} verbatim -> `wire/ILV.{h,cpp}` + new `wire/cmcomm.{h,cpp}`	`1f49d98` 2026-06-11; test_ilv ALL PASS
P0.c	[DONE] DONE	aamix.{h,c} verbatim -> `wire/AAMix.{h,cpp}` + `wire/resample.h` RESAMPLE ABI	`e0f2584`; [DONE] operator HL2 RX bench PASSED 2026-06-11
P0.d	[DONE] DONE	cmbuffs/cmaster/cmsetup verbatim -> `wire/CmBuffs`/`wire/CMaster`/`wire/cmsetup`; full `_cmaster` struct (PS surface); TxChannel carve-out DELETED	`afc7950` 2026-06-12; [DONE] **operator HL2 RX-regression bench PASSED 2026-06-12**
P1	[WIP] SHIPPED	obbuffs.c verbatim -> `wire/ObBuffs.{h,cpp}` (the TX-out seam; separate TU from cmbuffs; sendOutbound call DEFERRED to P2)	2026-06-12; dormant/wire-inert — optional operator RX smoke; clean build 0 warnings; mechanical diff IDENTICAL
P2	[DONE]	sendOutbound / sendProtocol1Samples fidelity audit of the dormant wire layer — **COMPLETE 2026-06-12** (P2.a eer completion + P2.b prn outbound surface verbatim + P2.c wire/Network.cpp sendOutbound verbatim, mechanical diff IDENTICAL; the ObBuffs ob_main hand-off restored). **P2.a SHIPPED 2026-06-12** — eer completion (verbatim struct + seam entries + the four CMaster restore sites); kills the null-`peer` landmine both reference functions deref. NEXT: P2.b prn outbound surface verbatim (outLRbufp/outIQbufp/OutBufp -> reference pointer fields + the HANDLE semaphore quartet; callers bend) -> P2.c wire/Network.cpp sendOutbound verbatim (P1 branch live, ETH branch DEFERRED) + restore the ObBuffs ob_main call.	Audit finding: Lyra's Step-14 OutboundRing/Ep2SendThread are functionally-parallel idiom translations (vector buffers, bool+cv semaphores) predating the verbatim mandate — retired at P4 when the verbatim chain goes live.
P3	[DONE]	netInterface registrations + obbuffs ring lifecycle. **SHIPPED + operator HL2 bench PASSED 2026-06-12** (RX fine; stop/start + shutdown clean) — create_rnet moved to once-per-process at the main.cpp QTimer AFTER create_xmtr (the reference C#-init order; fixes the per-open prn re-allocation leak); SendpOutboundTx(OutBound) restored at the reference site (netInterface.c:1761, tail of create_rnet); UpdateRadioProtocolSampleSize verbatim (netInterface.c:1836-1858, diff IDENTICAL) called per-open per StartAudio:45 — creates obbuffs rings 0/1 (2 new ob_main pump threads per session, idle: no producers until P4); destroy_obbuffs(0/1) at close() per StopAudio:112-113. RX side deliberately NOT registered (WdspEngine hybrid owns RX dispatch until P4 — the B.6.b clobber class).	[DONE] Operator HL2 bench PASSED 2026-06-12.
P4	[WIP]	Wire-LIVE switchover. **P4.a prep COMPLETE 2026-06-12** (2 wire-inert commits): sendProtocol1Samples ported verbatim into NEW src/wire/NetworkProto1.cpp (networkproto1.c:1204-1267, mechanical diff IDENTICAL; DORMANT — thread not started until P4.b) + io_keep_running global + the FrameComposer write_main_loop_hl2 tail flipped to the verbatim ReleaseSemaphore(hobbuffsRun[0/1]) pair; SetTXAPostGen{Run,Mode,ToneMag,ToneFreq,TTMag,TTFreq} wdspcalls entries (pre-cdef audit: gen.c:784-833 definition-site signatures, all 6 verified in the bundled wdsp.dll export table); TxReadBufp -> verbatim double* + create_rnet calloc (netInterface.c:1604) + Ep6RecvThread callers bent; hWriteThreadMain -> verbatim HANDLE.	NEXT = P4.b: the switchover, ONE commit, full HL2 bench gate (quartet + _beginthreadex at open(); dispatchAudioFrame asioOUT tee; Inbound(inid(1,0),...) mic feed; FSM TXA arming + TUN postgen; control-plane mapping; Step-14 translation retirements).

PureSignal is a committed feature — the P0.d `_cmaster` struct carries the full PS surface so v0.3 lands on top without a retrofit.

Historical stage index (pre-P0 lettered stages — superseded)

The original lettered plan below is retained for history. The P0 arc absorbed/superseded it: Stage C (ilv) = P0.b; Stage D (xcmaster) = P0.d; Stage E (TxChannel audit) = obsoleted by the P0.d TxChannel deletion; Stage B's idiom-translated AAMix was re-ported verbatim at P0.c.

Stage	Status	What	Notes
A	[DONE] DONE	CMaster.cpp wire-up stubs (Send*Outbound* + SetTCIRun)	Superseded by the P0.d full verbatim CMaster port.
B	[DONE] DONE	aamix.c direct port + wire into RX path	Shipped 2026-06-08 (idiom-translated); re-ported VERBATIM at P0.c.
C	[DONE] DONE (as P0.b)	ilv.c direct port (TX I/Q interleaver)	Landed verbatim at `wire/ILV.{h,cpp}`.
D	[DONE] DONE (as P0.d)	xcmaster pump body direct port	Landed verbatim in `wire/CMaster.cpp` (real stype/txid/chid + update[] CS).
E	[N/A] OBSOLETE	TxChannel reference-parity audit + reconciliation	TxChannel DELETED at P0.d — the verbatim create_xmtr replaces it.
F	[PEND] TBD	WDSP TXA chain components per CLAUDE.md §4.1 v0.2.1	compress / cfcomp / eqp / wcpagc — operator-driven per-feature direct port vs cffi-only call. Order TBD.
G	[PEND] TBD	PureSignal direct ports (v0.3)	iqc / calcc / xbuilder / delay. Lands when v0.3 PS work starts. Needs operator HL2+ with the PS hardware mod.
H	[PEND] TBD	TX/PS auto-attenuator FSM port	C# study only (PSForm.cs); Lyra-native rewrite.

Ordering note: the P0->P4 sequence above is the active linear plan. F–H land after P4 Wire-LIVE. The methodology (smallest revertable step + mechanical diff + operator bench gate) applies within each step.

OPEN QUESTIONS FOR OPERATOR

These are decisions that shape the plan but I cannot make unilaterally — surface them when picking up tomorrow:

1. **Stage C (ilv.c) vs Stage D (xcmaster) vs Stage E

(TxChannel audit) — which first?** All three are defensible:

- Stage C unlocks reference-faithful TX I/Q flow.
- Stage D removes the Lyra-native pump (cleaner end-state).
- Stage E formalizes what's already shipped (lowest risk,

highest confidence-gain).

2. **Do we re-port networkproto1.c + network.c + the Step 14

wire pieces direct from reference, OR leave them as the current mix of Lyra-native rewrites + original `hl2_stream.cpp`?** The Step 14 plan was midway through these when Stage B started. Direct-port would replace the rewrites with reference-verbatim ports. Operator call.

3. **WDSP TXA-chain components — direct port (~150-600 LOC

each) vs cffi-only?** The DLL exports work; cffi is faster to ship. Direct port gives Lyra-cpp full source visibility + lets us patch behavior at the source level. Per-component operator decision; default = cffi-only unless a real source- level need surfaces.

4. **PS work timing — start v0.3 (iqc/calcc/etc.) now in

parallel with TX, or sequence after TX is fully done?** The CLAUDE.md plan defers PS to v0.3 (after v0.2 TX release). Operator may want to start the ports early.

5. **The §15.27 "opt-in toggles that become defaults" cleanup

from the Python project — does it have a lyra-cpp analog here?** The Python project had a long backlog of legacy toggles to retire. Lyra-cpp doesn't carry the same legacy, but new direct-port commits may obsolete some Settings surfaces operator-knows-about — track per-component.

SHIPPED-WORK LOG

Newest at top. Cross-references SESSION_LOG.md for full arc details.

2026-06-12 — P0.d cmbuffs/cmaster/cmsetup VERBATIM port ([DONE] HL2 bench PASSED)

• **P0.d** `afc7950` on `tx-rebuild`: NEW `wire/cmsetup.{h,cpp}`

(cmMAX* 16/4/4/2/32 + the 8 id helpers + SetRadioStructure + set_cmdefault_rates); `wire/CmBuffs.{h,cpp}` rewritten verbatim (`cmb`, `*CMB`, malloc0/_aligned_free; Phase-C retrofit deviations removed); `wire/CMaster.{h,cpp}` rewritten verbatim (FULL `_cmaster` struct incl. the PureSignal surface, `enum AudioCODEC`, raw TCI fn ptrs, verbatim create/destroy_cmaster + create/destroy_xmtr + xcmaster + all Sendp*/Set* setters).

• **TxChannel RAII carve-out DELETED** (`src/wdsp/TxChannel.{h,cpp}`

retired) — the verbatim create_xmtr opens the WDSP TXA channel + TX analyzer (disp 1) + out[0..2] + ILV itself through the wdspcalls seam.

• main.cpp: SetRadioStructure(2,1,1,1,0,…) config before

create_cmaster (chid(0,0)=0 RX1 / chid(1,0)=1 TX = the live layout); create_xmtr() post-resolve_wdsp_calls (documented DEFERRED-CALLSITE accommodation); handler-1.5 = gated destroy_xmtr().

• Verification: clean build ZERO warnings; test_ilv ALL PASS;

mechanical diffs IDENTICAL (CmBuffs.{h,cpp} / cmsetup.h / CMaster.h / all 10 fully-verbatim cmaster.c functions) + live-line-subset PASS for the partially-deferred bodies (sole accommodation: `(char*)""` at XCreateAnalyzer).

• New at startup: TWO cm_main pump threads (streams 0+1; stream 0

idles — no Inbound producer); TX stays wire-quiescent (OutboundTx null until P3).

• [DONE] **GATE PASSED 2026-06-12: operator HL2 RX-regression bench**

("RX still working") — P1 (obbuffs.c) is unblocked.

2026-06-09 -> 2026-06-11 — P0.a/P0.b/P0.c verbatim-rewrite arc

• Operator rejected the 2026-06-09 "rule-#8" retrofit deviations

("I said PORT!") -> every ported file rewritten VERBATIM.

• **P0.a** `e0c0f4a`: `wire/wdspcalls.{h,cpp}` — the single

approved WDSP linkage seam (fn-ptr table, exact export names, PureSignal entry family complete + dumpbin-verified). `scratch/_typedef_fidelity_test.cpp` disproved the claimed "C++ collisions" (twin typedef / `complex[2]` / raw fn ptr all compile verbatim).

• **P0.b** `1f49d98`: ilv.{h,c} verbatim -> `wire/ILV.{h,cpp}`;

new `wire/cmcomm.{h,cpp}` (complex/PORT/malloc0/PI); pilv[] bank gone; SendpOutboundTx raw fn ptr (register AFTER create_xmtr — no null guards, reference ordering). Mechanical diff whitespace-only; test_ilv ALL PASS.

• **P0.c** `e0f2584`: aamix.{h,c} verbatim -> `wire/AAMix.{h,cpp}`

(supersedes the Stage-B idiom translation); `wire/resample.h` = public RESAMPLE ABI; wdspcalls retyped + flush_resample; WdspEngine outbound lambda -> static fn + TU-scope context ptr. [DONE] **Operator HL2 RX bench PASSED 2026-06-11.**

2026-06-08 EOD — Stage B aamix.c port COMPLETE

• **Stage B (aamix.c)** [DONE] direct port shipped + bench-validated

• Commits `27ac2fa` -> `7a50b07` on `tx-rebuild`

• Operator HL2+ bench 19:27:49 -> 19:28:47: audible RX through

both AK4951 jack and PC-Soundcard sinks; mute/unmute via dispatch; output-device swap; AGC mode changes; chain out=N tracks 1:1 with dispatch calls=N across both sinks

• B.6.b root cause: Stage-A SendpOutboundRx no-op stub at

`RadioNet.cpp:272` clobbered `aaMix_->Outbound` ~1 ms after `WdspEngine::openRx1` wired the real lambda. Fixed by deleting the stub (commit `8b8e0da`).

• 2-stage diagnostic instrumentation (chain @ `225518c` +

dispatch @ `12369bc`) localized the defect to a single line of code on the 2nd bench — pure-speculation cost avoided.

• Methodology lessons locked into this doc § "LOCKED

METHODOLOGY" above (counters-first when symptom is "nothing happens"; operator-empirical rule held; "do as Thetis does" continues to pay).

• Tasks #130 (parent) + #132 (Stage B.6.b) closed.

• Backup `_backups/lyra-cpp-2026-06-08-aamix-stage-B-COMPLETE-with-docs.bundle`.

Earlier work

See `SESSION_LOG.md` for the full session-by-session log. The Step 14 wire-layer arc (Stages 1, 1.5, 2a, 2b1, 2b2) is covered in detail there + in `EXECUTION_PLAN.md`'s progress- tracking sections. TX-1 components 2-6 (`main` branch) are covered in `project_lyra_cpp_tx.md` (memory file).

RESUME POINTER (next session)

First actions (locked discipline):

1. `git fetch origin`

2. `git status` + `git log --oneline -5 tx-rebuild origin/main`

3. Read this doc + the "READ FIRST" block in

`project_lyra_cpp_tx.md`

4. Read `SESSION_LOG.md` newest entry for arc detail

Then: wait for operator decision on which Stage (C / D / E / F / open-questions item) to land next. Each Stage gets its own grounded design + reference-read + smallest-revertable- step + bench-gated implementation pattern matching the Stage B arc.

Last updated: 2026-06-08 EOD — Stage B aamix.c port shipped + bench-validated. Next port target = operator's call.